

Python Power-Ups Placement Handbook

Chapter 1: PY1.1 – Comprehensions, Lambda & LEGB

This is the first chapter of a placement-oriented handbook based on your KMIT Talent Accelerator syllabus.

List Comprehensions

Syntax: [expression for item in iterable if condition]

```
nums=[1, 2, 3, 4]
squares=[x*x for x in nums]
```

Dictionary Comprehensions

Syntax: {key:value for item in iterable}

```
d={x:x*x for x in range(5)}
```

Set Comprehensions

Syntax: {expression for item in iterable}

Lambda Functions

```
square=lambda x:x*x
```

LEGB Rule

Local → Enclosing → Global → Built-in

Interview Tips

- Prefer comprehensions when building collections.
- Keep lambda expressions simple.
- Remember LEGB during scope debugging.

Revision Cheat Sheet

List: [expr for x in data if cond]

Dict: {k:v for x in data}

Set: {expr for x in data}

Lambda: lambda args: expression

LEGB: Local → Enclosing → Global → Built-in